NALAMMESH: OFFLINE-FIRST RURAL HEALTHCARE PLATFORM

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that **Mr./Ms. THARUN S**, bearing **Register Number 111624102222**, of **3rd Year, 5th Semester, Department of Computer Science and Engineering**, has successfully completed the mini project entitled **" NALAMMESH: OFFLINE-FIRST RURAL HEALTHCARE PLATFORM "** during the academic year **2024–2028** under my guidance and supervision.

The work submitted in this report is a bonafide record of the work carried out by the student/team as part of the academic requirements of the course.

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ABSTRACT

NalamMesh is an offline-first digital public healthcare platform designed for rural primary health care, where Sub-Centres, Primary Health Centres (PHCs), Community Health Centres (CHCs) and District Hospitals must work together although network connectivity is weak or absent. In many rural areas patient details are still written in paper registers, referral slips travel with the patient, and the receiving hospital learns about a critical case only when the ambulance arrives. Such methods cause delays in emergency care, loss of patient history and no clear way for district officers to see how referrals are handled.

The application lets health workers such as the Auxiliary Nurse Midwife (ANM), the Medical Officer (MO) and the District Health Officer (DHO) sign in with a staff ID and PIN and work according to their role. A health worker registers a patient, records vital signs and danger signs, and receives a triage result of RED, YELLOW or GREEN from an on-device TensorFlow.js model backed by a rule engine based on IPHS standards, which can only raise urgency and never lower it. A referral can then be sent to a higher facility. Records are stored on the device in IndexedDB, so the app keeps working without a network; uploads wait in a durable outbox and are sent when a connection returns. A Socket.io mesh relay carries live referrals between facilities, and a FastAPI district record service shows the receiving hospital a Pre-Arrival Board before the patient reaches the gate.

The system is developed as a responsive progressive web application that also runs as an Android app, with a relay server and a district service on SQLite or PostgreSQL. Automated testing confirmed that sign-in, role-based access, triage, referral hand-off, offline operation and record restoration work as expected. The system reduces delay in emergency referrals, keeps a complete and audited patient record, and helps the health department deliver continuous and accountable care to rural communities.

TABLE OF CONTENTS

S.No	Section	Page No.
1	Introduction	5
2	Objectives	5
3	Tools and Technologies	5
4	Architecture Diagram	6
5	Methodology	6
6	Results	7
7	Conclusion	10
8	Reference	10

1. INTRODUCTION

Rural public healthcare in India is delivered through a chain of facilities: the Sub-Centre in the village, the Primary Health Centre, the Community Health Centre and the District Hospital. Patients regularly need to move up this chain, for example a pregnant woman with danger signs, a child with severe malnutrition or a snakebite victim. In most districts these cases are recorded in paper registers, referrals are written on slips of paper, and the receiving facility is informed, if at all, by a phone call.

These traditional methods have several drawbacks. The patient's history is lost between facilities, there is no proof that a referral was sent or received, the receiving hospital cannot prepare a bed or a team in advance, and district officers have no live view of which emergencies are waiting. Internet connectivity in these areas is also unreliable, so ordinary online systems stop working exactly where they are most needed.

NalamMesh is an offline-first platform developed to overcome these limitations. It runs on the health worker's phone or computer and stores every record on the device first. Staff sign in with a PIN, register patients, record vitals and danger signs, receive a decision-support triage result, and send referrals to the next facility. When a connection is available, records and referrals reach the receiving facility through a mesh relay and a district record service, so the hospital sees the patient on its Pre-Arrival Board before the ambulance arrives.

The name NalamMesh combines "Nalam", meaning well-being, with "mesh", the network of facilities it connects. Its purpose is to make sure that every patient's record and referral travels with them across the continuum of care, building speed, safety and accountability into rural public health services.

2. OBJECTIVES

The main aim of this mini project is to design and develop a reliable, secure and offline-capable system for patient registration, triage and referral across rural health facilities. The specific objectives are:

- To provide a single application that works fully offline and synchronises automatically when a network is available.
- To allow secure sign-in with a staff ID and PIN, with role-based access for six roles such as ANM, Medical Officer and DHO.
- To capture complete patient details such as demographics, ABHA ID, vital signs, danger signs and chief complaint.
- To give a RED, YELLOW or GREEN triage result using an on-device AI model and IPHS clinical rules that can only raise urgency.
- To send referrals between Sub-Centres, PHCs, CHCs and District Hospitals and track their status until discharge.
- To show the receiving facility a Pre-Arrival Board so that it can prepare before the patient arrives.
- To record every access to identified patient data in an access log that the District Health Officer can review.
- To reduce paperwork and delays and improve continuity and accountability of rural healthcare.

3. TOOLS AND TECHNOLOGIES

The following tools and technologies were used to design, develop and test the NalamMesh platform.

Component	Technology	Purpose
Front end	Next.js 15, React 19, TypeScript, Tailwind CSS	Responsive user interface for sign-in, dashboards, OPD triage, referrals and boards
State and storage	Zustand, IndexedDB (idb)	App state and offline-first storage of patients, queue, referrals and stock
AI triage	TensorFlow.js, IPHS rule engine	On-device RED / YELLOW / GREEN triage with danger-sign rules
Mesh relay	Node.js, Socket.io	PIN sign-in, signed session tokens and live referral sync between facilities

Component	Technology	Purpose
Back end	Python FastAPI, SQLAlchemy	District record service, pre-arrival hand-off, access log and record restore
Database	SQLite, PostgreSQL 16	Stores uploaded patient records, referrals and the access log
Offline and mobile	Service worker, Capacitor	Installable app that opens with no network; Android build
Testing	Playwright, axe-core, custom verify suites	Browser end-to-end, offline, accessibility and unit checks
Development tools	Visual Studio Code, Git, GitHub Actions	Coding, version control and continuous integration
Browser	Google Chrome	Execution, debugging and testing of the web application

3.1 Hardware and Software Requirements

- **Hardware:** Processor: Intel Core i3 or above; RAM: 4 GB or more; Storage: minimum 1 GB free space; an Android phone (Android 8 or above) for the mobile app.
- **Software:** Windows, Linux or macOS with Node.js 20, Python 3.11, PostgreSQL or SQLite and a modern web browser.

4. ARCHITECTURE DIAGRAM

The system follows an offline-first, device-centred architecture. The health worker uses the app on a phone or computer, where every record is first saved in IndexedDB. The app talks to the mesh relay over WebSocket for sign-in and live referrals, and uploads records to the district record service over HTTPS when a connection is available. The receiving facility sees incoming patients on its Pre-Arrival Board, and the district service stores records in the database and logs every access.

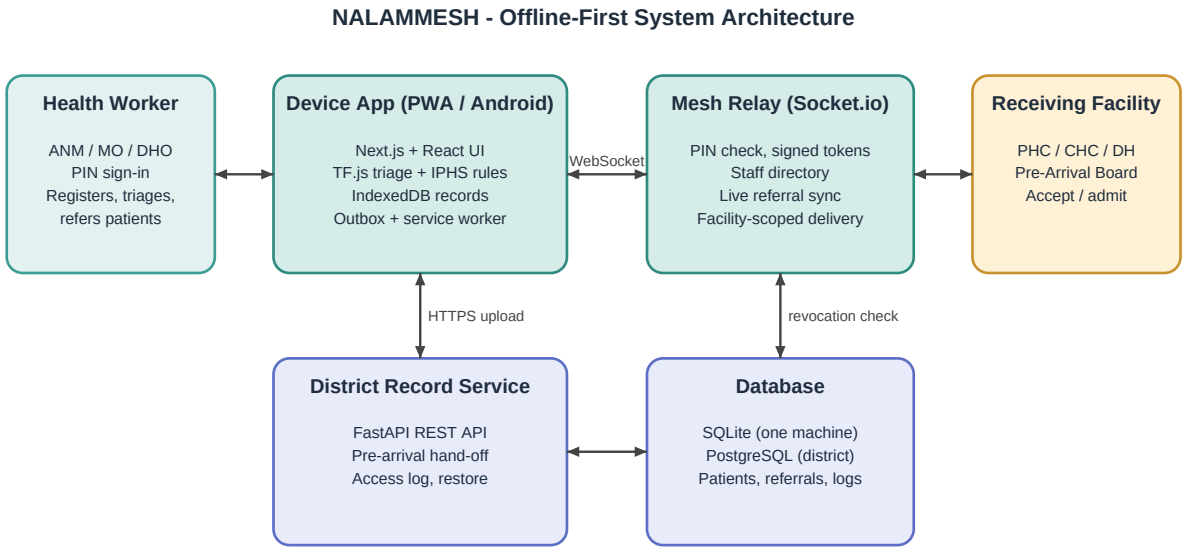


Figure 4.1: Architecture of the NalamMesh platform

4.1 Components

- **Health worker:** The ANM, Medical Officer or DHO signs in, registers and triages patients, and sends or receives referrals.
- **Device app:** Browser or Android app with the user interface, the on-device triage engine, IndexedDB storage and the upload outbox.
- **Mesh relay:** Checks PINs, issues signed session tokens, holds the staff directory and delivers referrals only to the facilities concerned.

- **District record service and database:** Stores uploaded records, serves the Pre-Arrival Board, restores a replacement phone and keeps the access log.
- **Receiving facility:** The PHC, CHC or District Hospital accepts the referral, prepares a bed and admits or discharges the patient.

5. METHODOLOGY

The project was developed using an incremental approach in which each module was designed, implemented and tested before moving to the next.

5.1 Requirement Analysis

The problems of the existing paper-based registration and referral process were studied and the required features were listed: offline operation, role-based sign-in, patient registration, triage, referrals, pre-arrival board and audit.

5.2 Design

A government-style interface was designed with a header, a top navigation bar and a sidebar listing each role's modules such as OPD Intake and Triage, Referrals, Pre-Arrival Board and Audit Trail. The OPD form was arranged in numbered sections: Patient Demographics and ABHA, and Physiological Vitals and Clinical Examination.

5.3 Implementation

- **Authentication module:** Staff sign in with a staff ID and PIN checked by the relay. Five wrong tries lock the account for five minutes and sessions use signed 12-hour tokens.
- **Registration and triage module:** The OPD form collects demographics, ABHA ID, vitals and danger signs. The TF.js model and IPHS rules give a RED, YELLOW or GREEN result and a slip states that the clinician decides.
- **Referral module:** A referral to a higher facility carries the clinical summary and moves through Sent, Acknowledged, Accepted, Arrived and Discharged; unacknowledged emergencies escalate to the DHO.
- **Offline sync module:** Every change is queued in a durable outbox and uploaded oldest first when online; the service worker lets the app open with no network.
- **Pre-Arrival Board:** The receiving facility reads patients en route from the district service, most urgent first.
- **Audit module:** Every request for identified records is logged with user, role, facility and status, and shown to the DHO on the Audit screen.

5.4 Referral Life Cycle

Step	Status	Description
1	Sent	Health worker refers a triaged patient to a higher facility; the record is queued for upload
2	Acknowledged	Receiving facility has seen the referral; unacknowledged emergencies escalate to the DHO
3	Accepted	Bed is held and the patient appears on the Pre-Arrival Board while in transit
4	Arrived / Admitted	Patient reaches the facility and is admitted for treatment
5	Discharged	Treatment is complete and the referral is closed

5.5 Testing

Each module was tested with valid and invalid inputs. 38 automated verify suites and browser checks covered sign-in, permissions, triage cases, referral flow, offline start, accessibility and a full production flow.

6. RESULTS

The NalamMesh application was implemented successfully and all modules worked as expected. The output screens of the system are shown below.

6.1 Staff Login

The Healthcare Staff Portal sign-in page lets a staff member choose the role, posting and staff name and enter the PIN. In the evaluation build, demo accounts for each of the six roles are listed on the right.

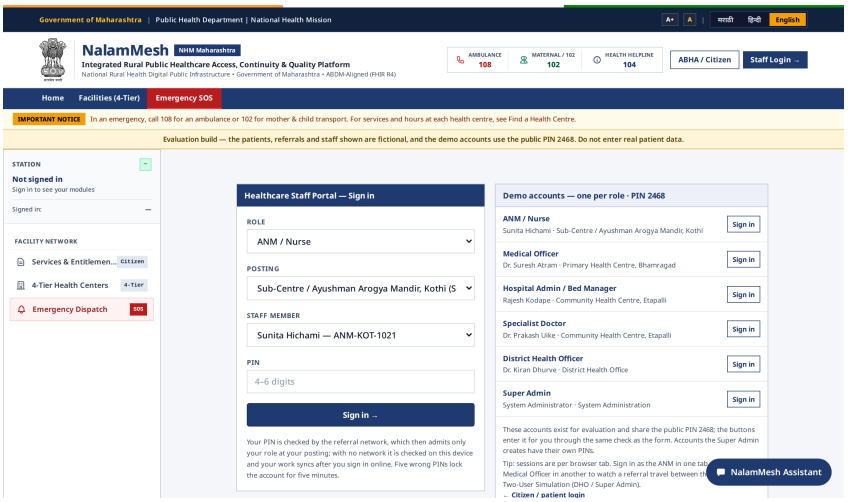


Figure 6.1: Staff Login page of NalamMesh

6.2 District Command Centre

After sign-in, the District Health Officer sees the District Health Command with referral counts, pending emergencies, escalation alerts, bed occupancy and a live map of the facility network.

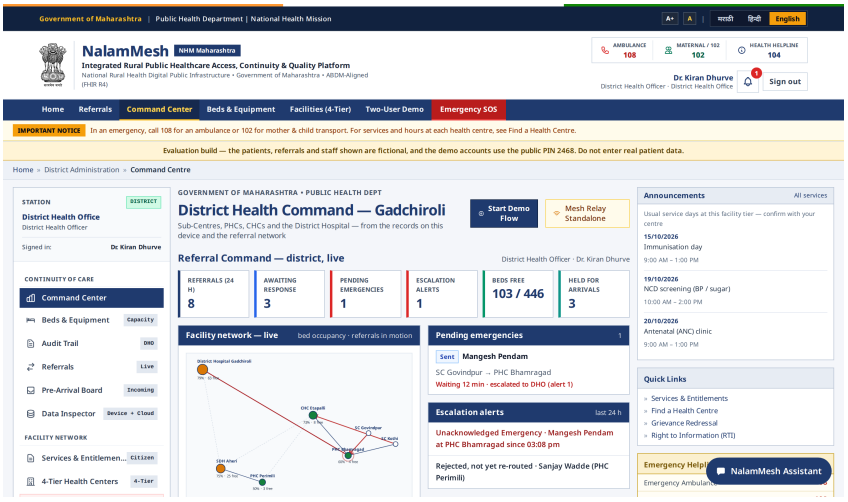


Figure 6.2: District Command Centre for the District Health Officer

6.3 OPD Patient Registration and Digital Triage

The OPD page collects the patient's demographics, ABHA ID, village, pregnancy or child status and the vital signs. The triage panel then shows the RED, YELLOW or GREEN result, and the live OPD queue board lists waiting patients.

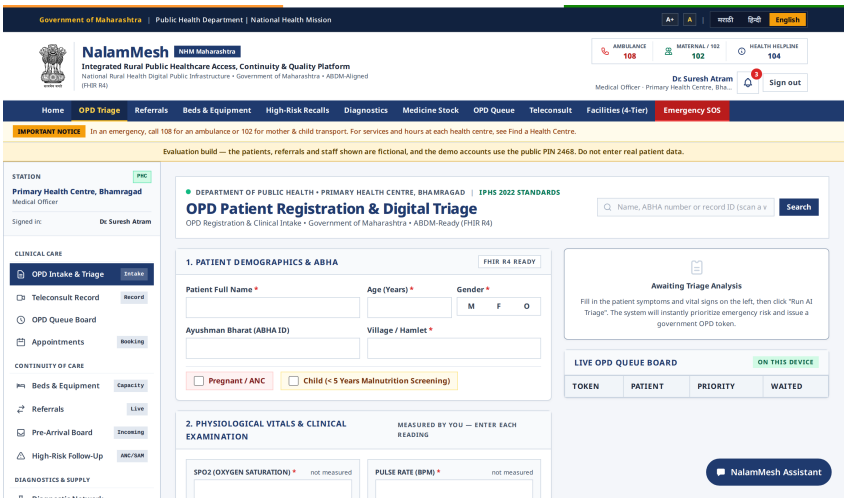


Figure 6.3: OPD Patient Registration and Digital Triage

6.4 Referrals between Facilities

The Referrals page shows incoming and outgoing referrals in columns for Awaiting response, Accepted, At receiving facility and Closed, with urgency, patient details and the time since each referral was sent.

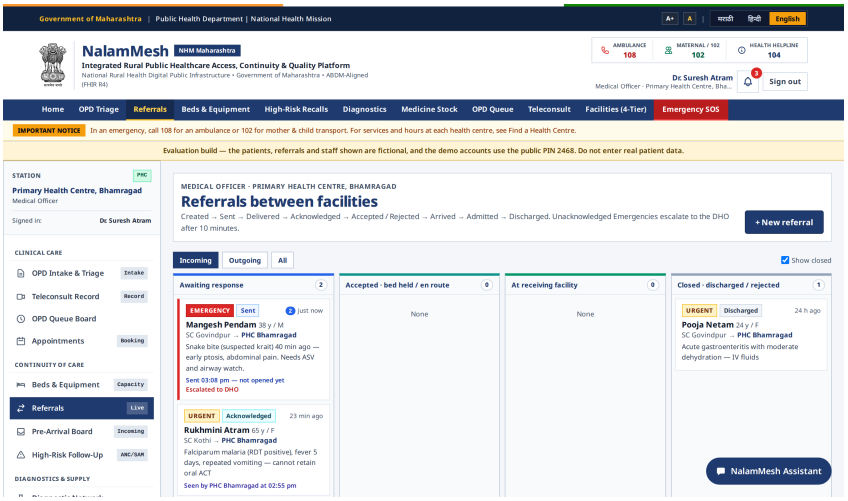


Figure 6.4: Referrals board showing referral status between facilities

6.5 ANM Dashboard

The ANM dashboard shows patients registered today, referrals awaiting reply, actions needed and follow-ups due, so the health worker can track each patient she has referred.

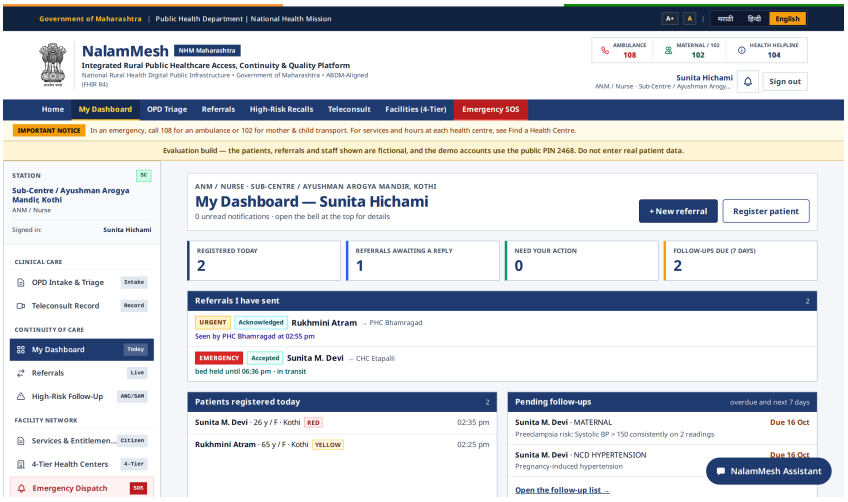


Figure 6.5: ANM dashboard with referrals sent and pending follow-ups

6.6 Test Results

S.No	Test case	Expected result	Result
1	Sign in with staff ID and PIN	Role dashboard opens; five wrong PINs lock the account	Pass
2	Register patient with danger signs	Triage result is RED and the slip is shown	Pass
3	Open a screen without permission	Access is refused for that role	Pass
4	Send referral to a higher facility	Receiving MO sees patient on Pre-Arrival Board	Pass
5	Use the installed app with server stopped	All ANM screens open; uploads wait in outbox	Pass
6	DHO opens the Audit screen	Every record access is listed in the access log	Pass
7	Sign in on a replacement phone	Facility records are restored from the district service	Pass
8	Run axe-core accessibility check	No WCAG 2.1 A/AA violation on 28 pages	Pass

7. CONCLUSION

NalamMesh provides a reliable, offline-first and accountable way to register, triage and refer patients across rural health facilities. By replacing paper registers and referral slips with a single application that works without a network, the system ensures that every patient's record is captured with complete details, travels with the referral and reaches the receiving hospital before the patient does. Features such as decision-support triage, the Pre-Arrival Board and the access log make the system practical and safe for health workers.

The project met its objectives and all tested modules worked correctly. It can help the health department respond faster to emergencies, improve continuity of care and strengthen accountability across the district.

7.1 Future Enhancements

- Clinical validation of the triage engine on real presentations and review of every threshold by clinicians.
- Full ABDM integration with ABHA verification and FHIR record exchange in production.
- Translation of all screens into regional languages through Bhashini, reviewed by clinicians.
- Recording of per-patient consent and notice under the Digital Personal Data Protection Act, 2023.
- District reports sent to the de-identified reporting service for analytics on referrals and outcomes.

8. REFERENCE

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